so adds extra complexity, so we recommend going with one of the available runtimes.

Next, we will compare AWS Lambda with EC2 virtual machines.

6.1.3 Comparing AWS Lambda with virtual machines (Amazon EC2)

What is the difference between using AWS Lambda and virtual machines? First is the granularity of virtualization. Virtual machines provide a full operating system for running one or multiple applications. In contrast, AWS Lambda offers an execution environment for a single function, a small part of an application.

Furthermore, Amazon EC2 offers virtual machines as a service, but you are responsible for operating them in a secure, scalable, and highly available way. Doing so requires you to put a substantial amount of effort into maintenance. In contrast, when building with Lambda, AWS manages the underlying infrastructure for you and provides a production-ready infrastructure.

Beyond that, AWS Lambda is billed per execution, and not per second like when a virtual machine is running. You don’t have to pay for unused resources that are waiting for requests or tasks. For example, running a script to check the health of a website every five minutes on a virtual machine would cost you about $3.71 per month. Executing the same health check with AWS Lambda will cost about $0.04 per month.

Table 6.1 compares AWS Lambda and virtual machines in detail. You’ll find a discussion of AWS Lambda’s limitations at the end of the